Jaewan Park

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Research Interests

Agentic AI, Foundation Models for Scientific Discovery

- **Towards Autonomous AI Scientists:** Building multi-agent systems for PDEs and in Life Science
- **Scalability of PDE Foundation models:** Understanding how pretraining data and objectives shape representations, developing efficient pretraining & adaptation strategies for generalization across tasks and domains.

Education

University of Illinois Urbana-Champaign, US *Aug 2022 – Present*
PhD student in Mechanical Science and Engineering
Focus: Generative AI, Foundation Models, and Representation Learning in PDEs

- GPA: 3.64 / 4.0
- **Coursework:** Graph Machine Learning, Diffusion Flow Matching Models, Statistical Learning Theory

Pohang University of Science and Technology (POSTECH), South Korea *Mar 2016 – June 2020*
B.S. in Mechanical Engineering - Summa Cum Laude (Rank : 1)

- GPA: 4.02 / 4.3 (3.89 / 4.0)

Stanford University, CA, US *June 2019 – Aug 2019*
International Honors Student

- GPA: 3.914 / 4.0

Publications

[1] J. Yoo, K. Kobayashi, **J. Park**, S. Puppala, S. Chakraborty, and S. B. Alam (2026). "Independent Latents, Robust Neural Operators." *NeurIPS 2026*.

[2] **J. Park**, J. Yoo, K. Kobayashi, S. Koric, S. B. Alam, I. Jasiuk, S. Chakraborty, and D. Abueidda (2026). "Stabilizing Autoregressive PDE Foundation Model Rollouts with Event-Triggered Context Healing." *SC26 proceedings*. arXiv:2609.32321.

[3] K. Kobayashi, F. Ahmed, **J. Park**, S. Sarkar, S. Koric, S. Chakraborty, and S. B. Alam (2026). "Virtual sensing to enable real-time monitoring of inaccessible locations & unmeasurable parameters." *Nature Communications*. doi:10.1038/s41467-026-77463-7.

[4] **J. Park**, F. Ahmed, K. Kobayashi, S. Koric, S. B. Alam, I. Jasiuk, and D. Abueidda (2026). "Learning What the Operator Misses: Residual Operator-Diffusion for Spatiotemporal PDE Solution Predictions." *npj Artificial Intelligence*. arXiv:2507.06133.

[5] **J. Park**, K. Kobayashi, Q. Liu, D. Abueidda, S. Koric, and S. B. Alam (2026). "Network architecture follows coupling in multiphysics systems: single vs. multiple branches in DeepONet and S-DeepONet." *Communications Engineering*. doi:10.1038/s44172-026-00714-4.

[6] **J. Park**, K. Kobayashi, Q. Liu, A. A. Koric, D. Abueidda, S. B. Alam, and S. Koric (2026). "Sequential deep operator neural networks for plastic and thermo-viscoplastic transient material behavior." *International Journal of Plasticity*, **201**, 104689. doi:10.1016/j.ijplas.2026.104689.

[7] **J. Park**, S. Kushwaha, J. He, S. Koric, Q. Liu, I. Jasiuk, and D. Abueidda (2026). "Nonlinear Inverse Design of Mechanical Multi-Material Metamaterials Enabled by Video Denoising Diffusion and Structure Identifier." *Engineering Applications of Artificial Intelligence*, **172**, 114368. doi:10.1016/j.engappai.2026.114368.

[8] **J. Park** (2026). "Transformer based neural operators for real-time accurate prediction." *NCSA student research conference*, Urbana-Champaign, IL.

[9] **J. Park** (2025). "Multi-material lattice design with diffusion models." *NCSA student research conference*, Urbana-Champaign, IL.

[10] J. He, S. Kushwaha, **J. Park**, S. Koric, D. Abueidda, and I. Jasiuk (2024). "Predictions of transient vector solution fields with sequential deep operator network." *Acta Mechanica*, **235**, 5257–5272. doi:10.1007/s00707-024-03991-2.

[11] S. Kushwaha*, **J. Park***, S. Koric, J. He, I. Jasiuk, and D. Abueidda (2024). "Advanced deep operator networks to predict multiphysics solution fields in materials processing and additive manufacturing." *Additive Manufacturing*, **88**, 104266. doi:10.1016/j.addma.2024.104266. *Co-first authors.

[12] **J. Park** (2024). "Multiphysics with operator networks: Towards Additive Manufacturing." *NCSA student research conference*, Urbana-Champaign, IL.

[13] J. He, S. Kushwaha, **J. Park**, S. Koric, D. Abueidda, and I. Jasiuk (2024). "Sequential deep operator networks (s-deeponet) for predicting full-field solutions under time-dependent loads." *Engineering Applications of Artificial Intelligence*, **127** (Part A), 107258. doi:10.1016/j.engappai.2023.107258.

[14] J. He, S. Koric, S. Kushwaha, **J. Park**, D. Abueidda, and I. Jasiuk (2023). "Novel DeepONet architecture to predict stresses in elastoplastic structures with variable complex geometries and loads." *Computer Methods in Applied Mechanics and Engineering*, **415**, 116277. doi:10.1016/j.cma.2023.116277.

Manuscripts Under Review

Where Grounding Ends: Boundary-Aware Reasoning for Biological Perturbation Prediction *Oct 2026*

DH Kwark, S Luo, ***J Park***, K Wang, A Abdrabou, AA Khan, V Kindratenko
Submitted

A Systematic Study of Attention in Data-Driven and Physics-Informed Deep Operator Networks for Thermal and Fluid Transport *Oct 2026*

AA Koric, Q Liu, S Roy, ***J Park***, K Kobayashi, S Koric, DW Abueidda, SB Alam
Submitted

FlowMesh: Deterministic Generative Transport of Physical Fields on Unstructured Meshes *Oct 2026*

S Verma, ***J Park***, J Yoo, K Kobayashi, S Chakraborty, SB Alam
Submitted

Ongoing Projects

Aligning Scientific Code Generation through Multi-Agents *In preparation*

- Building multi-agent framework for scientific code generation, aligning the process with human FEA/CFD scientist.
- Established a data-generation pipeline and created new data (problem definitions and code) for multiple types of libraries (FEM, FVM, Spectral, Discontinuous Galerkin), covering multiple languages (Python, C++, Julia).

Scaling Laws for Pretraining of PDE Foundation Models *In preparation*

- Investigating scaling laws and representation geometry to develop pretraining strategies that optimize data mixtures and compute allocation for adaptation to unseen PDEs.

Generative Style Transfer for Histopathology with Retrieval-Augmented Flow Matching *In preparation*

- Unpaired H&E-to-IHC translation via Vision Language Model and optimal transport using Histopathology Foundation Models, achieving high-fidelity generation with semantic consistency.

Technical Skills & Systems (Selected)

ML & Optimization: PyTorch, Tensorflow, Hugging Face, LoRA/QLoRA, vLLM
Scientific Computing: FEniCSx, Abaqus, Ansys, Dedalus
Experimentation: Hydra-based automation, Weights & Biases, Git
Systems & HPC: Linux, Bash, SLURM (job arrays), CUDA/cuDNN, conda/mamba, Docker/Apptainer
Language: Python, C, C++, R

Teaching & Mentoring

Teaching Assistant — TAM 210/211 (Statics) *Urbana–Champaign, IL*
Jan 2023 – May 2023